

Learning-Rate Calibration for LoRA-GRPO Across Verifiable Domains

Anonymous ACL submission

Abstract

Optimization settings for reinforcement learning with verifiable rewards (RLVR) are often transferred across tasks together with the algorithm. We test whether this practice is reliable for LoRA-based Group Relative Policy Optimization (GRPO). Using a shared backbone, adapter configuration, source-prompt pools, and evaluation pipeline, we compare conservative and higher-learning-rate GRPO with supervised or filtered self-training across four standard verifiable domains; a 31-example legal task is reported separately as a small-data stress test. Under our implementation, a conservative learning rate produces little or inconsistent held-out improvement, whereas higher-rate runs improve Math, Science, Medicine, and Commonsense, with substantial variation across domains and training runs. Law is a boundary case rather than evidence of universal superiority. Training diagnostics are consistent with an under-update hypothesis, but do not establish that learning rate changes GRPO’s divergence direction or creates new knowledge. Exploratory transfer and reward-variance analyses further identify where the effect is less reliable. Our results support a scoped conclusion: LoRA-GRPO optimization strength must be calibrated for the model, data scale, and reward structure rather than inherited as a fixed cross-domain recipe.

1 Introduction

Reinforcement learning with verifiable rewards (RLVR) has become a widely used post-training approach for language-model reasoning. DeepSeekMath and DeepSeek-R1 (Shao et al., 2024; DeepSeek-AI, 2025) established Group Relative Policy Optimization (GRPO) as a practical alternative to actor-critic methods, and DAPO (Yu et al., 2025) refined its data and optimization pipeline. Related recipes are now applied to medicine (Lai et al., 2025),

information extraction (Anonymous, 2025), and other verifiable tasks.

This expansion raises a calibration question. A learning rate is not an algorithm-independent constant: its effect depends on update parameterization, adapter capacity, batch construction, data scale, and the distribution of reward signal. In particular, a rate reported for full-parameter training is not automatically appropriate for LoRA. Existing cross-paper comparisons rarely isolate these factors because models, data, and evaluation protocols change simultaneously.

We study this question for LoRA-GRPO using Qwen2.5-7B-Instruct under a shared adapter configuration and evaluation pipeline. The primary comparison covers Math, Science, Medicine, and Commonsense; Law, with only 31 training examples, is treated as a separate small-data stress test. We compare GRPO with supervised fine-tuning on GSM8K solutions and with rejection-sampling fine-tuning on filtered, self-generated correct responses in the non-math domains. An additional filtered self-training variant changes the response-selection rule and serves as a control, not as a distinct source of external supervision.

The controlled sweeps reveal a reproducible qualitative pattern in the saved runs: at a conservative rate of $5e-7$, LoRA-GRPO changes held-out accuracy little, while higher rates often produce larger improvements and $1e-4$ can collapse training. We use $5e-7$ as an intentionally conservative baseline; DeepSeekMath itself reports $1e-6$ for its policy model (Shao et al., 2024). Because learning rate is varied while the data and evaluation are fixed, the pattern identifies optimization strength as an important lever in this implementation. It does not rule out complementary explanations such as prompt overfitting, limited data diversity, or evaluation artifacts, which we treat explicitly as alternatives.

High-learning-rate spot checks on additional

checkpoints suggest that the effect is not unique to Qwen2.5. These runs do not, however, establish cross-model learning-rate sensitivity because most lack a matched conservative-rate control. We therefore distinguish evidence that a high-rate configuration can help from evidence that the same calibration curve transfers across model families.

Our contributions are as follows.

1. We provide a controlled multi-domain study of learning-rate sensitivity in LoRA-GRPO, with four primary domains and Law explicitly separated as a small-data stress test.
2. We show that conservative and higher-rate LoRA-GRPO occupy empirically different update regimes in this implementation: conservative runs often under-update, intermediate rates can improve held-out accuracy, and excessive rates can collapse.
3. We clarify the boundaries of this result through filtered-self-training controls, benchmark-level transfer analyses, additional-model spot checks, and an exploratory reward-variance diagnostic, while separating observed performance from mechanistic hypotheses.

2 Related Work

RLVR for LLM Reasoning. Post-training with reinforcement learning is increasingly used to improve LLM reasoning. DeepSeekMath, DeepSeek-R1, and GRPO (Shao et al., 2024; DeepSeek-AI, 2025) reduce reliance on demonstrations and a learned value model. DAPO (Yu et al., 2025) combines dynamic sampling with other optimization changes, including filtering groups with zero reward variance during training. Extensions to medicine (Lai et al., 2025), relation extraction (Anonymous, 2025), and event understanding (Gao et al., 2024) report gains within individual domains. Our focus is narrower than establishing whether RLVR works in general: we isolate how one LoRA-GRPO implementation responds to optimization strength across several verifiable domains.

SFT versus RL. The choice between token-level supervision and reward optimization affects both the information available during training and the resulting update. Chain-of-thought SFT (Wei et al., 2022) learns from complete target

sequences, whereas RLVR uses outcome feedback. Prior work studies differences in memorization, generalization, and trace distributions (Chu et al., 2025; Wang et al., 2026; Yue et al., 2025). In our non-math experiments, however, the supervised targets are themselves generated by the base model and filtered for correctness. We therefore describe this baseline as rejection-sampling fine-tuning (RFT), not as external-teacher supervision, and interpret the comparison as a recipe-level comparison rather than an equal-information comparison.

Training Dynamics and Domain Transfer.

Understanding failed post-training runs requires separating observations from explanations. The “Delay, Plateau, Collapse” pattern (Egashira et al., 2026), reward overoptimization (Gao et al., 2023), and diminishing returns from redundant supervised data (Zhou et al., 2024) provide relevant alternatives. A reward–test gap may arise from insufficient policy movement, training-prompt overfitting, limited data diversity, or evaluation mismatch. Our controlled learning-rate intervention is consistent with the first explanation under the studied setup, but does not exclude the others without additional diagnostics.

3 Experimental Setup

We design a controlled sensitivity study across four primary domains, two supervised/self-training baselines, and a LoRA-GRPO learning-rate sweep. Law is retained as a small-data stress test rather than pooled with the four primary domains.

3.1 Domain Selection

We select domains spanning procedural reasoning and knowledge-intensive multiple-choice tasks. Table 1 summarizes the benchmark composition. Science combines ARC-Challenge and ScienceQA; Commonsense combines ARC-Easy and HellaSwag. We report component-benchmark scores in addition to pooled accuracy because the component test sets differ substantially in size.¹

¹ScienceQA is evaluated text-only: images from its multimodal subset are discarded while retaining the question text and answer options. All base evaluations use greedy decoding and the same answer parser as the trained models.

Domain	Type	Train Source	Reward	N _{SFT}	N _{RL}	N _{te}	Test Set
Math	Procedural	GSM8K	Exact match	5,000	2,000	1,319	GSM8K
Science	Mixed	Science pool	MCQ match	3,000	2,000	5,413	ARC-C + ScienceQA
Medicine	Knowledge	MedQA	MCQ match	5,000	2,000	1,273	MedQA
Law	Knowledge	LegalBench	Binary	31	31	10,403	LegalBench
Commonsense	Mixed	Commonsense pool	MCQ match	5,000	2,000	12,418	ARC-E + HellaSwag

Table 1: Domain overview. N_{SFT} is the maximum supervised/filtered-self-training data size swept; N_{RL} is the GRPO prompt budget. The methods draw from the same source-prompt pool, but filtering changes the number of retained supervised examples. Law is a micro-data stress test ($N=31$), not a like-for-like domain comparison. Benchmarks: GSM8K (Cobbe et al., 2021), ARC (Clark et al., 2018), MedQA (Jin et al., 2021), and LegalBench (Guha et al., 2024).

3.2 Training Methods

Our primary experiments apply all methods to Qwen2.5-7B-Instruct (Qwen Team, 2024) with LoRA (Hu et al., 2022) (rank 64, alpha 128). We additionally report high-rate GRPO spot checks on Qwen3-8B (Yang et al., 2025), Mistral-7B-Instruct-v0.3, Yi-1.5-9B-Chat, and DeepSeek-7B-chat. These checkpoints use the same adapter targets and evaluation pipeline, with model-specific tokenizers and chat templates. Because most additional checkpoints do not include a matched low-rate control, they test whether the high-rate configuration can improve performance, not whether the full learning-rate sensitivity curve transfers.

LoRA-GRPO (RLVR). We use GRPO (Shao et al., 2024) with binary reward: exact match for Math, option-label match for Science, Medicine, and Commonsense, and binary/categorical matching for Law. For each prompt, we sample $K=8$ completions and use $\beta=0.001$ KL regularization. The primary domains use $N=2000$ prompts; Law has 31 unique prompts. We sweep $5e-7$, $5e-6$, $2e-5$, and $1e-4$. We call $5e-7$ our *conservative LoRA baseline*; it is not the policy learning rate reported by DeepSeekMath, which uses $1e-6$ in a different full-parameter setup.

Supervised FT and rejection-sampling FT (RFT). Math uses the provided GSM8K solutions. For Science, Medicine, Law, and Commonsense, we sample eight responses from the same base model and retain the shortest correct response as the supervised target. These non-math runs are therefore RFT rather than external-teacher SFT. We vary the retained-data budget from $N = 50$ to $N = 5000$, subject to the number of prompts for which sampling produces a correct response.

Filtered self-training control (FST; previously “OPD”). We sample $K=8$ responses at temperature 0.7, retain the first correct response

for each covered prompt, and fine-tune with the same LoRA configuration. This is a second rejection-sampling/self-training construction, not knowledge distillation from an external teacher. Relative to RFT, it changes the response-selection rule and the retained training set; consequently, it is an informative control but does not isolate the effect of self-generation by itself.

SFT/RFT→GRPO. A sequential pipeline first applies token-level fine-tuning and then GRPO. We vary the preceding supervised-data amount ($N=100$ vs. $N=5000$). These hybrid results are diagnostic because some conditions have only one training run.

3.3 Fairness Protocol

The credibility of our comparison rests on controlling confounds.

- **Same base model** across all methods and domains.
- **Same LoRA configuration:** rank 64, alpha 128, applied to all linear layers.
- **Same source-prompt pools;** supervised filtering may retain fewer prompts than GRPO.
- **Same evaluation:** greedy decoding (temperature 0), identical prompt template.
- **Comparable nominal epochs:** approximately three epochs per condition. GRPO additionally generates $K = 8$ responses online, so the methods are not compute-matched.

The comparison is intentionally recipe-level rather than equal-information. Supervised/RFT runs receive token-level targets, whereas GRPO receives only outcome rewards and incurs online

rollout cost. We therefore do not attribute differences solely to the optimization objective. A complete accounting of generated tokens, retained examples, optimizer updates, and GPU-hours is needed for a compute-normalized comparison.

All experiments were conducted on NVIDIA H200 GPUs (143GB). Appendix 13 reports the available aggregate compute inventory; we avoid using the raw number of exploratory runs as evidence strength because seed counts differ by condition.

Full hyperparameters are reported in Appendix B.

4 Results

4.1 Conservative LoRA-GRPO Often Under-Updates

Under the conservative 5e-7 setting, the saved LoRA-GRPO runs provide little or inconsistent improvement over the base model (Table 2). This observation is specific to our adapter-based implementation and should not be read as a failed reproduction of full-parameter DeepSeekMath.

The effect is not identical across domains. Math and Science show small changes at 5e-7 and larger gains at the higher-rate point. Medicine slightly degrades at 5e-7 and improves at 2e-5, but with appreciable run-to-run variability. Commonsense has the largest mean increase and the largest variability. Law is different: its higher-rate mean is close to RFT, reinforcing its role as a small-data boundary case rather than evidence for a universal rule.

Because the data and evaluation are held fixed within each sweep, the contrast is consistent with an under-update hypothesis: 5e-7 may produce insufficient effective movement for this LoRA parameterization. The sweep alone does not exclude training-prompt overfitting, limited data diversity, or answer-parser effects. We therefore treat policy movement and robustness measurements as diagnostics rather than proof of a unique mechanism.

Filtered self-training is a weak control, not a separate signal source. The alternative filtered self-training construction changes response selection and retained-example counts relative to RFT. Its mean gains are below one point on Science and Medicine (Table 2), but this does not show that “external signal is necessary”: both constructions use gold-answer filtering, and their trace

length and data size are not matched. A controlled selection-rule ablation is needed before assigning the gap to the learning objective.

We next examine how the observed performance changes along the learning-rate sweep.

4.2 Learning Rate Changes the LoRA-GRPO Regime

Medicine provides the clearest sensitivity example in the saved artifacts. Increasing the learning rate improves the reported mean through 2e-5, whereas 1e-4 collapses. This establishes a useful intervention pattern under fixed data, but not a universal optimum.

Across the evaluated points, the mean increases from 5e-7 to 5e-6 and 2e-5 (Table 3). This is evidence that the conservative LoRA update is too weak under this configuration. It does not by itself establish latent-knowledge consolidation: benchmark exposure, training-prompt overlap, response-format learning, and evaluation sensitivity remain plausible contributors.

We do not include the previous reward-trajectory figure because its plotted arrays are not recoverable from the sanitized run logs. The revised diagnostic will be generated directly from trainer logs and will report reward, policy-reference KL, update or gradient norm, and clipping statistics with explicitly defined uncertainty.

The appropriate conclusion is therefore implementation-scoped: LoRA-GRPO should not inherit a learning rate without validation for its model, adapter, and data regime.

Law illustrates the boundary. With only 31 training prompts, 2e-5 hurts relative to 5e-6. This is compatible with small-data overfitting, but one task is insufficient to establish a dataset-size scaling law.

Learning rate changes update magnitude but does not change the mathematical direction of the divergence optimized by GRPO. We therefore avoid describing the low-rate regime as “pure reverse KL” or claiming that a larger rate captures new modes. The available evidence supports an empirical regime change, with its mechanism left as a hypothesis.

Method	Math	Science	Medicine	Law	Commonsense
Base (zero-shot)	84.4	71.1	59.2	57.6	43.8
Supervised/RFT (reported cfg.)	87.8±0.5 (4)	73.6±0.5 (4)	59.7±0.9 (4)	58.8±0.1 (3)	46.8±0.3 (4)
GRPO (5e-7)	85.0±0.5 (2)	71.5±0.4 (2)	58.7±0.5 (2)	54.6 (1)	44.4 (1)
GRPO (higher-rate point)	92.0±0.6 (4)	79.1±1.5 (4)	64.4±2.9 (4)	58.6±0.9 (4)	59.8±6.7 (3)
Filtered self-training ctrl.	–	71.7±0.2 (5)	60.0±0.9 (4)	–	–
Δ higher-rate vs base	+7.6	+8.0	+5.2	+1.0	+16.0

Table 2: Descriptive accuracy (%) reconstructed from saved evaluation artifacts. Parentheses give the number of saved runs and error terms are standard deviations across those runs; training-seed provenance must be verified before treating them as independent replications. The higher-rate point is 2e-5 except for Law (5e-6). It improves four primary domains but has high variability on Commonsense; Law remains comparable to RFT. Component-benchmark and dev-selected results are required for the final comparison.

GRPO Learning Rate	Medicine (%)	Δ
Base	59.2	–
lr = 5e-7 (2 runs)	58.7±0.5	−0.4
lr = 5e-6 (1 run)	61.3	+2.1
lr = 2e-5 (4 runs)	64.4±2.9	+5.2
lr = 1e-4	collapsed	–

Table 3: Descriptive LoRA-GRPO learning-rate sweep on Medicine. The number of saved runs differs by condition. Performance improves from the conservative setting to 2e-5, but variability at 2e-5 is non-negligible; 1e-4 collapses. A dev-selected, balanced-seed comparison is required for an unbiased headline result.

Takeaways

1. In this LoRA-GRPO implementation, 5e-7 produces little or inconsistent held-out improvement.
2. Intermediate learning rates improve four primary domains in the saved runs, while variability and the Law stress test rule out a universal setting.
3. Filtered self-training is a related baseline whose selection rule, retained data, and compute must be matched before drawing objective-level conclusions.

4.3 SFT Data Scaling: The Inverted-U

The supervised/RFT sweeps are domain dependent (Appendix F). Math displays an inverted-U over the evaluated data sizes, Science improves gradually, and Medicine is approximately flat on average. These curves describe associations; they do not show that additional demonstrations overwrite reasoning or that token-level supervision cannot add medical knowledge. Dataset composition, filtering rate, response length, and benchmark exposure all vary with domain and may explain part of the pattern. The previous scaling figure included interpolated points and is therefore removed until it can be regenerated from run-level artifacts only.

Domain	Base	PPL	Peak Δ	N^*
Math	84.4%	4.34	+3.4%	100
Science	71.1%	2.92	+2.5%	5000+
Medicine	59.2%	2.46	+0.5%	flat

Table 4: Exploratory comparison of training-data perplexity and the observed supervised/RFT scaling pattern. The three points are insufficient for correlation or threshold claims and are confounded by dataset, response construction, and task format.

4.4 Exploratory Association with Training-Data Perplexity

We additionally compute the mean per-token cross-entropy of each supervised training set under the base model. With only three domain-level observations, this analysis is exploratory and cannot validate perplexity as a predictor.

The ordering in Table 4 is compatible with a distribution-mismatch explanation, but Science and Medicine are close in perplexity and their training data differ in many other ways. We therefore do not use these three observations to prescribe a data budget.

4.5 An Exploratory Reward-Variance Diagnostic

frac_reward_zero_std measures the fraction of prompts for which all $K = 8$ sampled responses receive the same binary reward. Such groups have zero within-group advantage variance under the sampled batch. The statistic is closely related to the zero-variance groups handled online by DAPO-style dynamic sampling (Yu et al., 2025); here we examine it only as a pre-training descriptor.

Medicine has a lower reported fraction than Math, so more sampled groups contain reward contrast. This observation is consistent with

Transfer	Method	Acc (%)	Std
Math → Science	GRPO	82.6	1.7
	SFT	84.2	0.6
	FST	75.8	–
Med → Science	GRPO	87.8	1.2
	SFT	77.9	0.5
Sci → Medicine	GRPO	77.4	0.8
	SFT	70.6	0.8
	FST	67.3	0.2

Table 5: Exploratory cross-domain transfer. Each row trains on the source domain and evaluates on the target benchmark. GRPO is higher on two pairs and RFT is higher on Math→Science. Standard deviations summarize the saved evaluations where available; the table does not control for source-data format or evaluation-prompt effects.

greater opportunity for group-relative updates. It does not establish that the resulting changes are new knowledge or rule out benchmark memorization.

Math has a higher reported fraction and weaker cross-benchmark transfer to MATH-500. This is compatible with prompt- or format-specific improvement, although benchmark difficulty and contamination are alternative explanations.

Three domains cannot validate predictive thresholds such as 0.4 or 0.6. We therefore present the statistic as a hypothesis-generating descriptor. Its practical value beyond online signal-aware filtering requires a larger model-domain grid and a controlled low-rate filtering baseline. The previous phase-diagram figure is removed because the underlying KL points are not recoverable from the sanitized logs.

4.6 Cross-Domain Transfer

We also report cross-domain evaluation as an exploratory robustness probe. Because the source and target benchmarks differ in format, topic, and base accuracy, these comparisons do not isolate a learning mechanism.

GRPO is higher on Medicine→Science and Science→Medicine, whereas RFT is higher on Math→Science (Table 5). The mixed result is useful evidence against a universal-superiority claim. It motivates, but does not prove, the hypothesis that the source domain and response format determine which learned behavior transfers.

5 Analysis and Discussion

5.1 Under-Update Hypothesis and Alternative Explanations

Learning rate scales the optimizer update but does not turn GRPO from one KL direction into another. Our primary interpretation is therefore operational: under the studied LoRA parameterization, $5e-7$ may induce too little effective policy movement, while intermediate rates allow a larger update and $1e-4$ is unstable. This interpretation should be tested with real policy-reference KL, update norms, clipping fractions, and matched alternatives that change the number of updates rather than the learning rate.

Several alternatives remain. A higher rate may fit training-prompt regularities faster; old benchmarks may overlap pretraining data; option-label behavior may change without deeper reasoning; and the filtered-self-training baselines differ in retained examples and trace length. The present results do not distinguish latent-knowledge consolidation from more reliable expression of previously encountered benchmark patterns.

Reward variance and learning rate are complementary quantities: the former indicates whether a sampled group contains contrast, while the latter affects the size of the resulting optimizer step. Signal-aware filtering could improve the efficiency of low-rate training, so a direct low-LR filtering comparison is necessary before describing learning rate as the sole lever.

The hybrid results are likewise descriptive. A short supervised stage followed by GRPO performs differently from a long supervised stage, but the current design simultaneously changes checkpoint quality, response diversity, and the data seen before RL. It therefore cannot attribute the difference to forward- versus reverse-KL geometry.

5.2 What Does Higher-Rate GRPO Change?

Cross-benchmark transfer provides a partial probe. Math shows a small reported change on MATH-500, whereas Medicine shows a larger change on MMLU-Medical (Appendix M). These patterns suggest that the learned changes differ by source domain, but they do not identify new knowledge: benchmark contamination, prompt mismatch, and different target difficulty remain uncontrolled.

Prior theory explains why zero-variance

groups contribute no group-relative signal (Bae et al., 2025), and prior empirical work questions whether RLVR expands the solvable frontier (Yue et al., 2025). Our experiments connect these questions to learning-rate sensitivity, but do not extend either result theoretically. Even when held-out accuracy is flat, the procedure remains reinforcement learning; the relevant question is whether its updates produce useful generalization.

5.3 Practical Recipe Framework

We report high-rate GRPO spot checks on four additional checkpoints from three other organizations (Table 16). Positive changes at 2e-5 show that the configuration is not useful only for Qwen2.5, but matched low-rate controls and repeated training runs are required before claiming cross-model sensitivity. We therefore avoid a universal practitioner decision rule.

6 Conclusion

This study identifies learning rate as an important calibration variable for LoRA-GRPO. A conservative setting yields little or inconsistent improvement in the saved runs, intermediate rates improve four primary domains, and an excessive rate can collapse training. The effect is not universal: Law is a small-data boundary case, Commonsense is highly variable, and cross-domain transfer is mixed. These findings support calibrating optimization strength on held-out development data rather than importing a fixed recipe. They do not establish a new information-theoretic mechanism or knowledge acquisition.

Limitations

Our evidence is limited to 7–9B instruction-tuned checkpoints and LoRA updates. A low full-parameter learning rate can be inappropriate for LoRA, so full-parameter controls are necessary before generalizing to GRPO as an algorithm. Most non-math tasks are multiple choice, and open-ended verifiable tasks remain untested. Several benchmarks are old enough that pretraining contamination is plausible; controlled comparisons share this exposure, but contamination limits claims about latent or newly acquired knowledge. The current saved runs also have uneven replication, and earlier sweeps selected configurations using test performance. Final claims require

independent seed verification, dev-only selection, benchmark-level macro aggregation, and paired uncertainty estimates. Finally, the non-math supervised baselines use gold-filtered self-generated responses rather than an external teacher, and are not yet matched to the alternative filtered-self-training control in retained examples, trace length, or generated-token budget.

References

- Anonymous. 2025. [R1-re: Cross-domain relation extraction with rlvr](#). *arXiv preprint arXiv:2507.04642*.
- Sanghwan Bae and 1 others. 2025. Online difficulty filtering for reasoning oriented reinforcement learning. *arXiv preprint arXiv:2504.03380*.
- Tianzhe Chu and 1 others. 2025. Sft memorizes, rl generalizes: A comparative study of foundation model post-training. *arXiv preprint arXiv:2501.17161*.
- Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord. 2018. Think you have solved question answering? try arc, the ai2 reasoning challenge. *arXiv preprint arXiv:1803.05457*.
- Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, and 1 others. 2021. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*.
- DeepSeek-AI. 2025. Deepseek-rl: Incentivizing reasoning capability in llms via reinforcement learning. *arXiv preprint arXiv:2501.12948*.
- Kazuki Egashira, Mark Vero, Jasper Dekoninck, Florian E. Dorner, Robin Staab, and Martin Vechev. 2026. [Delay, plateau, or collapse: Evaluating the impact of systematic verification error on rlvr](#). *arXiv preprint arXiv:2605.02909*.
- Jun Gao, Huan Zhao, Changlong Rui, and 1 others. 2024. [Eventrl: Enhancing event extraction with outcome supervision](#). *arXiv preprint arXiv:2402.11430*.
- Leo Gao, John Schulman, and Jacob Hilton. 2023. Scaling laws for reward model overoptimization. In *International Conference on Machine Learning*.
- Neel Guha, Julian Nyarko, Daniel E Ho, Christopher Ré, Adam Chilton, Ines Chami, and 1 others. 2024. Legalbench: A collaboratively built benchmark for measuring legal reasoning in large language models. *Advances in Neural Information Processing Systems*.

Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. Lora: Low-rank adaptation of large language models. *arXiv preprint arXiv:2106.09685*.

Di Jin, Eileen Pan, Nassim Oufattole, Wei-Hung Weng, Hanyi Fang, and Peter Szolovits. 2021. What disease does this patient have? a large-scale open domain question answering dataset from medical exams. *Applied Sciences*, 11(14).

Jiaxin Lai and 1 others. 2025. Med-rlvr: Emerging medical reasoning from a 3b base model via reinforcement learning. *arXiv preprint arXiv:2502.19655*.

Qwen Team. 2024. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*.

Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Mingchuan Zhang, Y.K. Li, Y. Wu, and Daya Guo. 2024. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*.

Ziming Wang and 1 others. 2026. Rl squeezes, sft expands: Understanding the topology of post-training reasoning. In *International Conference on Learning Representations*.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc V Le, and Denny Zhou. 2022. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems*, volume 35.

An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, and 1 others. 2025. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*.

Qiyang Yu and 1 others. 2025. Dapo: An open-source llm reinforcement learning system. *arXiv preprint arXiv:2503.14476*.

Yang Yue, Zhiqi Chen, Rui Lu, Andrew Zhao, Zhaokai Guan, Zhixuan Li, Flemming Hvidberg Liu, Tianyu Peng, Zichen Shen, Guang Chen, and 1 others. 2025. Does reinforcement learning really incentivize reasoning capacity in LLMs beyond the base model? *arXiv preprint arXiv:2504.13837*.

Chunting Zhou, Pengfei Liu, Puxin Xu, Srinu Iyer, Jiao Sun, Yuning Mao, Xuezhe Ma, Avia Efrat, Ping Yu, Lili Yu, and 1 others. 2024. LIMA: Less is more for alignment. In *Advances in Neural Information Processing Systems*, volume 36.

A Practical Reporting Guide

The present evidence does not support universal thresholds based on base accuracy, perplexity, or

frac_reward_zero_std. A defensible workflow is instead to (i) identify the update parameterization, including LoRA rank and target modules; (ii) select learning rate and checkpoint on a held-out development set; (iii) inspect reward contrast and real policy-movement diagnostics; (iv) report component-benchmark and macro scores with independent training seeds; and (v) test answer-parser and option-order sensitivity for MCQ tasks. High-rate settings should be treated as candidates to validate, not defaults.

B Hyperparameters

Parameter	GRPO	SFT
Base model	Qwen2.5-7B-Instruct	
LoRA rank	64	64
LoRA alpha	128	128
Target modules	All linear	All linear
Learning rate	5e-7 to 2e-5	2e-5
Batch size	2 × 4 accum	4 × 4 accum
Training epochs	~3 (via steps)	3
Num. generations	8	–
Temperature (train)	1.0	–
KL coefficient	0.001	–
Max new tokens	2048 / 1024	–

Table 6: Training hyperparameters for LoRA-GRPO and supervised/RFT runs. We use 5e-7 as a conservative LoRA baseline and evaluate larger rates as sensitivity points. The sweep does not establish a universal optimum.

C SFT Data Scaling Details

Full SFT accuracy curves for Math and Medicine across all data sizes:

N	Math (%)	Medicine (%)
50	87.4	–
100	88.6	59.5
200	86.9	–
500	78.6	59.5
1000	–	58.6
1500	–	60.2
2000	–	61.7
5000	75.9	57.8

Table 7: SFT accuracy at different data sizes. Math peaks at $N=100$ with steep degradation beyond. Medicine peaks at $N=2000$ with gentler slopes. Base accuracy: Math 84.4%, Medicine 59.2%.

Math shows an inverted-U over the evaluated sizes: accuracy rises by 4.2 points at $N=100$ and falls below base at $N=5000$. Medicine is flatter on average. These domain-specific curves are

descriptive; they do not identify a phase transition or distinguish format learning from knowledge acquisition.

D Reward Function Details

Math. We extract the final numerical answer and compare against the ground truth after normalization (removing commas, converting fractions to decimals).

MCQ domains (Science, Medicine, Law, Commonsense). We extract the letter answer (A/B/C/D) from the model’s response and compare against the gold label. Reward is 1 for exact match, 0 otherwise.

E SFT Data Construction

For Science, Medicine, Law, and Commonsense, we construct demonstrations by rejection sampling from the base model. For each question, we sample eight responses at temperature 0.7 and keep the shortest response with the correct final answer; questions with no correct sample are excluded. This construction favors prompts already solvable by the base model and changes both coverage and response length. We therefore call it rejection-sampled fine-tuning (RFT), and do not treat it as a controlled teacher-versus-on-policy comparison.

For Math, we use the step-by-step solutions provided in GSM8K.

F Full SFT Data Scaling Results

The saved RFT/SFT sweeps are summarized in Tables 8–10. Math shows an inverted-U over the evaluated sizes, Medicine is approximately flat on average, and Science improves gradually. The grids and seed counts are uneven, so these patterns should be confirmed with balanced runs selected on a development set.

G GRPO Learning Rate Sweep Details

The saved GRPO learning-rate runs are summarized in Table 11. A higher rate is associated with larger gains on Math, Science, and Medicine, whereas the 31-example Law stress test is best at 5e-6. Because learning rate, domain, and dataset size are not crossed factorially, these runs do not establish dataset size as the cause of the different preferred settings.

For GRPO at 5e-7, two saved runs are available for Math (84.7%, 85.4%) and Science (71.2%,

71.7%). Their means differ from base by less than one point. Training configurations must confirm that these are independent seeds before formal uncertainty reporting.

H Cross-Model Validation: Qwen3-8B

We additionally evaluate the high-rate configuration on Qwen3-8B. These spot checks test whether the setting can improve another checkpoint; without a matched low-rate sweep, they do not establish that the same learning-rate effect transfers. Results are in Table 12.

At 2e-5, the reported Qwen3 checkpoint scores exceed its supervised/RFT scores on these three tasks (Table 12). Because these are mostly single-run configurations without a matched low-rate Qwen3 control, they are evidence for a useful high-rate setting, not for a shared calibration curve or a headroom mechanism.

I Training Dynamics: Reward Trajectories

The previous draft reported summary reward and KL values without retaining the underlying step-level logs in the sanitized artifact. We therefore omit those values from the evidentiary argument. A replacement diagnostic should be generated directly from raw trainer logs and define how each statistic and uncertainty band is computed.

Category	Models	GPU-hrs	%
SFT (all domains)	55	22	17%
GRPO standard lr	15	18	14%
GRPO lr sweep	10	12	9%
GRPO lr=2e-5 (seeds)	10	14	11%
SFT→GRPO hybrids	4	5	4%
Cross-model (Qwen3)	10	12	9%
Cross-model (Mistral/Yi/DS)	8	18	14%
KL/K ablation	6	8	6%
Evaluation (vLLM)	—	18	14%
Failed/exploratory	82	29	22%
Total recorded compute	—	~200	100%

Table 13: Approximate compute ledger, including exploratory and failed runs. The category counts are not a complete model inventory and should not be interpreted as independent evidence. Final reporting will be regenerated from run manifests.

J SFT→GRPO Hybrid Details

The full SFT→GRPO hybrid results on Math are in Table 14, varying the amount of preceding SFT data.

N	s42	s123	s456	Avg
50	87.4	–	–	87.4
100	87.8	88.6	87.6	87.8
200	86.9	86.3	85.4	86.2
500	78.6	79.1	79.8	79.2
2k	77.6	79.2	79.2	78.7
5k	75.9	77.6	77.8	77.1

Table 8: Math SFT (%). Base: 84.4. Peak at $N=100$, steep degradation beyond.

N	s42	s123	s456	Avg
100	59.5	60.0	60.6	59.7
500	59.5	58.3	60.9	59.3
1k	58.6	59.5	–	59.1
2k	61.7	58.5	59.5	59.4
3k	59.2	59.3	57.9	58.8
5k	58.7	–	58.5	58.6

Table 9: Medicine RFT (%). Base: 59.2. The mean is approximately flat and individual runs vary across N .

N	s42	s123	s456	Avg
100	71.7	72.5	71.6	71.9
500	71.8	73.6	71.9	72.4
1k	72.5	–	72.8	72.6
2k	73.4	73.1	72.7	73.1
5k	73.8	73.6	74.0	73.6

Table 10: Science RFT (%). Base: 71.1. The saved means increase across the evaluated sizes.

The saved runs show three descriptive scaling patterns: an inverted-U on Math, an approximately flat mean on Medicine, and gradual improvement on Science. These differences do not isolate a cause because the domains differ in data construction, filtering coverage, response length, and possible benchmark exposure.

Learning Rate	Math	Science	Medicine	Law	Pattern
Base	84.4	71.1	59.2	57.6	–
5e-7 (conservative)	85.0	71.5	58.7	54.5	Little/hurt
5e-6	–	73.9	61.3	58.9	Moderate
1e-5	–	–	–	58.8	–
2e-5	92.0	79.1	64.4	56.4	Larger/mixed
1e-4	–	–	0.0	–	Collapsed
Δ (higher-rate vs base)	+7.6	+8.0	+5.2	+1.0	

Table 11: Descriptive LoRA-GRPO accuracy (%) across saved learning-rate runs. Higher-rate means improve Math, Science, and Medicine; Law prefers the lower 5e-6 point in this small-data stress test. Uneven replication and test-based sweep selection preclude a universal optimum claim.

	Math	Science	Medicine
<i>Qwen3-8B</i>			
Base	81.5	73.9	50.5
SFT best	82.9 _{5k}	73.4 _{5k}	62.1 ₅₀₀
GRPO lr=2e-5	90.5	78.7	62.9
Δ best vs base	+9.0	+4.8	+12.4
<i>Qwen2.5-7B-Instruct</i>			
Base	84.4	71.1	59.2
SFT best	87.8 ₁₀₀	73.6 _{5k}	59.5 ₁₀₀
GRPO lr=2e-5	92.0	79.1	64.4
Δ best vs base	+7.6	+8.0	+5.2

Table 12: High-rate GRPO spot checks on two model generations. Values for Qwen3 are single saved checkpoints; Qwen2.5 GRPO values are saved-run means. Both tables compare against each model’s base and best reported supervised/RFT configuration, but only Qwen2.5 has a learning-rate sweep.

Pipeline	s42	s123	Avg	Δ
Base	84.4	84.4	84.4	–
SFT($n=100$) only	87.8	88.6	88.2	+3.8
GRPO lr=2e-5 only	92.8	–	92.8	+8.4
SFT($n=100$)→GRPO	89.8	88.7	89.3	+4.9
SFT($n=5000$)→GRPO	82.9	–	82.9	–1.5

Table 14: SFT→GRPO hybrid results on Math. Seed coverage is shown explicitly. The saved hybrid scores are below the saved GRPO-only run, but the comparison is not compute matched and the $N=5000$ hybrid has one run.

K KL Penalty and Group Size Ablation

We report exploratory single-configuration ablations of KL coefficient β and group size K on Medicine. The grid is incomplete and is not sufficient to identify learning rate as the primary variable independently of these interactions.

Within the high-rate seed-42 rows, accuracy decreases as β increases. This is compatible with stronger KL regularization limiting policy change, but the table lacks replicated cells and matched training diagnostics. We therefore do not infer a general tuning rule from this ablation.

The hybrid results are consistent with an interaction between preceding supervised training and subsequent GRPO, but they do not identify the mechanism. A direct test should match compute and record group-wise reward variance, response diversity, KL, and update norms before attributing the difference to reduced exploration.

Setting	Accuracy (%)	Δ vs base	Interpretation
lr=5e-7, $\beta=0.001$	58.8	−0.4	Saved run
lr=5e-7, $\beta=0$	57.0	−2.2	Saved run
lr=2e-5, $\beta=0.001$	65.4	+6.2	Seed 42
lr=2e-5, $\beta=0.01$	62.4	+3.2	Seed 42
lr=2e-5, $\beta=0.1$	60.2	+1.0	Seed 42
lr=2e-5, $K=4$	62.9	+3.7	Seed 42

Table 15: Exploratory KL-coefficient and group-size ablations on Medicine. Rows are saved single runs and do not form a complete learning-rate $\times$ KL $\times$ group-size design; differences may include seed variation.

L Cross-Model Validation

Gain magnitude does not simply increase as base accuracy decreases: for example, the saved Yi run gains less than the saved Mistral run. With one checkpoint per cell and no matched low-rate controls, this observation is descriptive and cannot be attributed to a specific uncertainty statistic.

is not free of external supervision. It also differs from RFT in response selection and retained examples; it is therefore not a clean self-generated-versus-teacher comparison.

M Out-of-Distribution Transfer

The three transfer ratios differ substantially, but they do not separate knowledge acquisition, format effects, benchmark difficulty, contamination, or evaluation noise. We retain this table as an exploratory robustness probe and reserve predictive claims for a larger preregistered model–domain grid with disjoint training and evaluation suites.

N Exploratory Reverse-Transfer Control

We also report a reverse-transfer control in which a math-supervised checkpoint is evaluated on other benchmarks. Because the training and evaluation templates differ and the provenance of the reported MATH-500 training split requires verification, this analysis is exploratory rather than a direct test of format versus knowledge.

The checkpoint is lower on the reported math evaluation and higher on the two MCQ evaluations. A plausible explanation is improved adherence to the answer format, but changes in prompting, benchmark composition, or checkpoint selection could produce the same pattern. A format-permutation evaluation and matched open-ended scoring are needed before using this result to support a mechanism claim.

O Filtered Self-Training Results

Filtered self-training (FST; called OPD in the previous draft) fine-tunes on the first correct output among $K=8$ samples at temperature 0.7. Correctness filtering uses gold labels, so this condition

Model	Lab	Base (%)	GRPO lr=2e-5 (%)	Δ	Observation
<i>Medicine (MedQA, 1273 test)</i>					
Qwen2.5-7B-Instruct	Alibaba	59.2	64.4	+5.2	Saved-run mean
Qwen3-8B	Alibaba	50.5	62.9	+12.4	Newer generation
Yi-1.5-9B-Chat	01.AI	43.3	49.3	+6.0	Llama architecture
Mistral-7B-Instruct	Mistral AI	34.3	42.8	+8.5	European model
DeepSeek-7B-chat	DeepSeek	12.1	20.3	+8.2	2023 model (weak base)
<i>Science suite (ARC-Challenge + ScienceQA, 5413 test)</i>					
Qwen2.5-7B-Instruct	Alibaba	71.1	79.1	+8.0	Primary model
Qwen3-8B	Alibaba	73.9	78.7	+4.8	Single run
Mistral-7B-Instruct	Mistral AI	61.2	70.4	+9.2	Single run

Table 16: High-rate GRPO spot checks on five model checkpoints from four organizations. The 2e-5 runs improve over each checkpoint’s base score in the reported tasks. Most rows are single-run and lack a matched conservative-rate control, so the table does not establish cross-model learning-rate sensitivity.

Domain	OOD Benchmark	Base	GRPO	Δ OOD	Δ ID	Transfer Ratio	frac_zero_std
Medicine	MMLU-Medical	69.4	78.0	+8.6	+5.2	$1.65\times$	0.2
Math	MATH-500	50.8	52.2	+1.4	+7.6	$0.18\times$	0.6
Commonsense	WinoGrande	69.2	73.0	+3.8	+16.0	$0.24\times$	0.7

Table 17: Exploratory transfer results from the saved evaluation artifacts. Transfer ratio is $\Delta\text{OOD}/\Delta\text{ID}$ and can exceed one because the ID and OOD suites have different scales and difficulty. The three rows are insufficient to validate frac_reward_zero_std as a predictor.

Benchmark	Base (%)	SFT-Math (%)	Δ
Math-500	60.2	42.4 $\pm$ 0.2	−17.8
ARC-Challenge	74.9	87.5 $\pm$ 0.1	+12.6
MMLU	61.1	68.2 $\pm$ 0.1	+7.1

Table 18: Exploratory evaluation of a math-supervised checkpoint on three benchmarks. The accuracy changes are compatible with response-format effects, but do not uniquely identify them. Three saved runs are reported.

	Science	Medicine
Base	71.1	59.2
FST	71.7 $\pm$ 0.2 (5)	60.0 $\pm$ 0.9 (4)
Δ	+0.6	+0.9
Filtering rate	69.1%	53.7%

Table 19: Filtered self-training results on Science and Medicine. Parentheses give the number of saved runs. The small mean changes are descriptive; a matched-prompt, matched-token RFT/FST experiment is required to attribute the difference to response provenance or selection.